MATH 1210 Worksheet 4 Summer 2023

Suggested exercises, more exercises can be found in the textbook or on Dr. R.S.D. Thomas' webpage for archived material:

`http://home.cc.umanitoba.ca/~thomas/Courses`

1. Write in simplified Cartesian form
$$\frac{(1-i)^{14}}{(\sqrt{3}+i)^{10}}.$$

Solution: We have $1-i=\sqrt{2}e^{-i\pi/4}$ and $\sqrt{3}+i=2e^{i\pi/6}$. Hence

$$\begin{aligned}\frac{(1-i)^{14}}{(\sqrt{3}+i)^{10}} &= \frac{(\sqrt{2}e^{-i\pi/4})^{14}}{(2e^{i\pi/6})^{10}}\\ &= \frac{2^7e^{-i14\pi/4}}{2^{10}e^{i10\pi/6}}\\ &= \frac{2^7e^{i\pi/2}}{2^{10}e^{-i\pi/3}}\\ &= \frac{1}{2^3}e^{i\pi(1/2+1/3)}\\ &= \frac{1}{2^3}e^{i5\pi/6}\\ &= \frac{1}{2^3}\left(-\frac{\sqrt{3}}{2}+\frac{i}{2}\right)\\ &= -\frac{\sqrt{3}}{16}+\frac{i}{16}.\end{aligned}$$

2. Find all solutions to $z^5=16(1-i)^2$. Write the solutions in exponential form and use principal values of the argument.

Solution: We have $(1-i)^2=1^2-2i+i^2=-2i$. Hence the equation can be rewritten as

$$z^5=16\cdot(-2i)=2^5e^{-i\pi/2}. \tag{1}$$

Let us now solve equation (1). Set $z=re^{i\theta}$. If z is a solution then

$$z^5=r^5e^{i5\theta}=2^5e^{-i\pi/2}.$$

This implies that $r=2$ and

$$5\theta=-\frac{\pi}{2}+2\pi k=\frac{\pi(4k-1)}{2}$$

for some integer k. Hence

$$\theta=\frac{\pi(4k-1)}{10}$$

and any solution z of the equation is of the form

$$z=2e^{i\frac{\pi(4k-1)}{10}}$$

for some integer k. For the values of $k = 0, 1, 2, 3, 4$ we find the five distinct solutions

$$\begin{aligned}
k &= 0 \qquad & z_0 &= 2e^{-i\frac{\pi}{10}} \\
k &= 1 \qquad & z_1 &= 2e^{i\frac{3\pi}{10}} \\
k &= 2 \qquad & z_2 &= 2e^{i\frac{7\pi}{10}} \\
k &= 3 \qquad & z_3 &= 2e^{i\frac{11\pi}{10}} = 2e^{-i\frac{9\pi}{10}} \\
k &= 4 \qquad & z_4 &= 2e^{i\frac{15\pi}{10}} = 2e^{-i\frac{5\pi}{10}}.
\end{aligned}$$

3. Consider the polynomial $P(x) = x^3 + 10x^2 - 37x + 26$.
   (a) Use Descartes rule of sign to determine the number of possible positive and negative real roots.
   (b) It is given to you that -13 is root. Find all other possible rational roots. Reduce the list as much as possible using (a).
   (c) Write $P(x)$ as a product of linear factors.

Solution:

(a) The polynomial $P(x)$ has 2 sign changes. Hence, by Descartes rule of sign it has 0 or 2 positive roots. The polynomial
$$P(-x) = -x^3 + 10x^2 + 37x + 26$$
has one sign change. Hence, by Descartes rule of sign, the polynomial $P(x)$ has one negative root.

(b) By the rational root theorem, if p/q is a rational root in reduced form, then p divides 26 and q divides 1. Hence the possible values of q are ± 1, the possible values for p are $\pm 1, \pm 2, \pm 13 \pm 26$ and the possible values for p/q are
$$\pm 1, \pm 2, \pm 13, \pm 26.$$
But we already know that -13 is a rational root, and that it has no other negative roots by part (a). Thus, the other possible values for p/q are
$$1, 2, 26.$$

(c) Write P as a product of linear factors. By trial and error (the result of part (b) helps) we find that 1 and 2 are roots of $P(x)$ since
$$\begin{aligned}
P(1) &= 1 + 10 - 37 + 26 = 0 \\
P(2) &= 2^3 + 10 \cdot 2^2 - 37 \cdot 2 + 26 = 8 + 40 - 74 + 26 = 0.
\end{aligned}$$
Hence the three roots of $P(x)$ are $1, 2, -13$. By the factor theorem we can factor $P(x)$ in the linear factors:
$$P(x) = (x-1)(x-2)(x+13).$$

4. Consider the polynomial
$$P(x) = -x^3 + \frac{2}{3}x^2 - 2x + \frac{3}{2}.$$
   (a) Use Bounds theorem to find an upper bound on the absolute values of the roots.
   (b) Show that there are no roots in the interval $[10, 100]$.

(c) Find a list of rational roots by using the rational root theorem. Use parts (a) and (b) to reduce the list. *Hint: clear the denominators.*

Solution:

(a) By Bounds theorem, if x is a root of $P(x)$ then

$$|x| < \frac{M}{|-1|} + 1 = 3$$

where $M = \max\{|\frac{2}{3}|, |-2|, |\frac{3}{2}|\} = 2$.

(b) By Descartes rule of signs, $P(x)$ has no negative real roots since

$$P(-x) = x^3 + \frac{2}{3}x^2 + 2x + \frac{3}{2}$$

has no sign changes in the coefficients. But by part (a) we also know that there are no roots such that $|x| \geq 3$. In particular, combining these two informations we find that there are no roots in the interval $[3, 100]$. In particular, there are no roots in the interval $[10, 100]$.

(c) To apply the rational root theorem, we need to clear the denominators. Indeed, note that p/q is a root of $P(x)$ if and only if p/q is a root of

$$6P(x) = -6x^3 + 4x - 12x + 9.$$

By the rational root theorem, p divides 9 and q divides -6. So the possible values for p are

$$\pm 1, \pm 3, \pm 9$$

and for q

$$\pm 1, \pm 2, \pm 3, \pm 6.$$

So a list of possible values roots p/q is

$$\pm 1, \pm 3, \pm 9, \pm 1/2, \pm 3/2, \pm 9/2, \pm 1/3, \pm 1/6.$$

By the previous parts we can remove the negative roots and the roots $\pm 9/2, \pm 9, \pm 3$ because their absolute values is greater than 3. The reduced list of possible values is

$$1/6, 1/3, 1/2, 1, 3/2.$$

5. Consider the matrices

$$A = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix} \qquad B = \begin{pmatrix} -2 & 4 \\ 1 & 0 \end{pmatrix}.$$

Verify that AB and BA are not the same, by computing the two products.

Solution: On the one hand we have

$$AB = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} -2 & 4 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -3 & 4 \\ -1 & 8 \end{pmatrix}.$$

On the other hand we have

$$BA = \begin{pmatrix} -2 & 4 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} 6 & 14 \\ 1 & -1 \end{pmatrix}.$$

6. Consider the matrices

$$A = \begin{pmatrix} 3 & 2 & 0 \\ -1 & 1 & -2 \\ 1 & 2 & -3 \end{pmatrix} \qquad B = \begin{pmatrix} 1 & 3 & 0 \\ 2 & 5 & -2 \end{pmatrix} \qquad C = \begin{pmatrix} 4 & -5 \\ 7 & 0 \\ 3 & -2 \\ 1 & -1 \end{pmatrix}$$

$$D = \begin{pmatrix} -1 & 2 & 1 & -2 \\ 0 & -1 & 1 & 0 \end{pmatrix} \qquad E = \begin{pmatrix} 1 & 3 \\ 2 & -2 \end{pmatrix}.$$

Evaluate the following expressions if they exist or explain why they are undefined.

(a) AB^TE

(b) BA^TE

(c) $A + C$

(d) $2DB^T - 3E$

(e) $2D - C^T$

Solution:

(a) First we mutliply A with B^T:

$$AB^T = \begin{pmatrix} 3 & 2 & 0 \\ -1 & 1 & -2 \\ 1 & 2 & -3 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 5 \\ 0 & -2 \end{pmatrix} = \begin{pmatrix} 9 & 16 \\ 2 & 7 \\ 7 & 18 \end{pmatrix}.$$

Hence

$$AB^TE = \begin{pmatrix} 9 & 16 \\ 2 & 7 \\ 7 & 18 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 2 & -2 \end{pmatrix} = \begin{pmatrix} 41 & -5 \\ 16 & -8 \\ 43 & -15 \end{pmatrix}$$

(b) B is 2×3 matrix and A is a 3×3 matrix. Hence, BA^T is a well defined 2×3 matrix. But BA^TE is not defined, since BA^T is a 2×3 and E a 2×2 matrix.

(c) The sum $A + C$ is not well defined since A is a 3×3 matrix and C is a 4×2 matrix.

(d) The matrix B^T is a 3×2 matrix, and D is a 2×4 matrix. Thus the product DB^T is not defined.

(e) We have

$$\begin{aligned} 2D - C^T &= 2 \cdot \begin{pmatrix} -1 & 2 & 1 & -2 \\ 0 & -1 & 1 & 0 \end{pmatrix} - \begin{pmatrix} 4 & 7 & 3 & 1 \\ -5 & 0 & 3 & 1 \end{pmatrix} \\ &= \begin{pmatrix} -2 & 4 & 2 & -4 \\ 0 & -2 & 2 & 0 \end{pmatrix} - \begin{pmatrix} 4 & 7 & 3 & 1 \\ -5 & 0 & 3 & 1 \end{pmatrix} \\ &= \begin{pmatrix} -6 & -3 & -1 & -5 \\ 5 & -2 & -1 & -1 \end{pmatrix} \end{aligned}$$

7. Find all the values t such that the matrix

$$A = t \cdot \begin{pmatrix} 2 & 1 \\ t+1 & 2 \end{pmatrix} - \begin{pmatrix} 0 & 2 \\ 1 & -1 \end{pmatrix}^T$$

is upper triangular but not diagonal.

Solution: We write the matrix as:

$$A = t \cdot \begin{pmatrix} 2 & 1 \\ t+1 & 2 \end{pmatrix} - \begin{pmatrix} 0 & 2 \\ 1 & -1 \end{pmatrix}^T = \begin{pmatrix} 2t & t \\ t^2+t & 2t \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 2 & -1 \end{pmatrix}$$
$$= \begin{pmatrix} 2t & t-1 \\ t^2+t-2 & 2t+1 \end{pmatrix}.$$

This 2×2 matrix is upper triangular if $t^2 + t - 2 = 0$, which has solutions $t = 1$ and $t = -2$. But when $t = 1$ the matrix is

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix},$$

hence diagonal. Hence $t = -2$ is the only value for which A is upper triangular but not diagonal. For this value we have

$$A = \begin{pmatrix} -4 & -3 \\ 0 & 3 \end{pmatrix}$$